

Analysis vs Analytics:

Analysis is a backward-looking activity that involves examining data to understand past events.

It involves dividing the dataset into manageable chunks, studying them individually, and examining their relationships with other parts

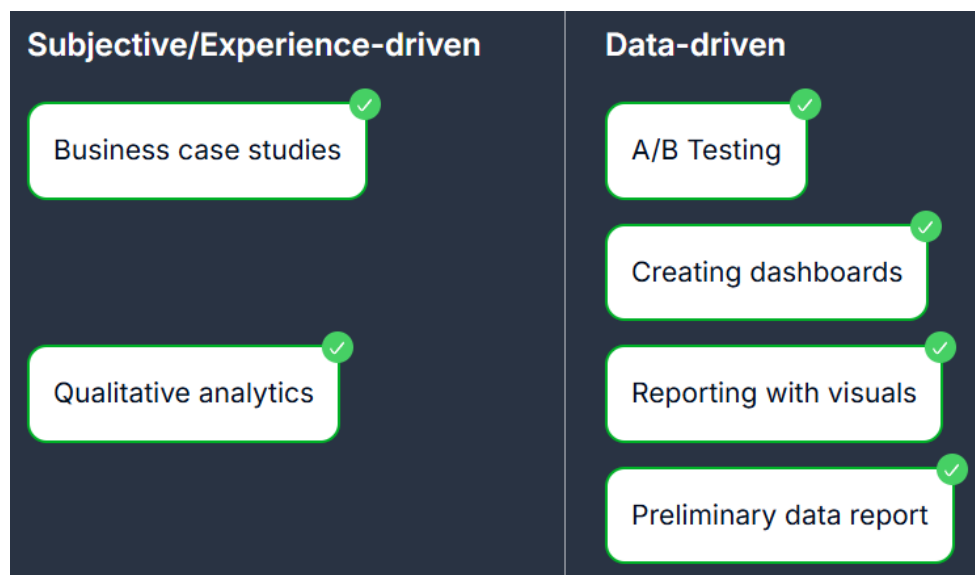
Analytics is a forward-looking activity that involves exploring potential future events.

It involves applying logical and computational reasoning to the components derived from an analysis.

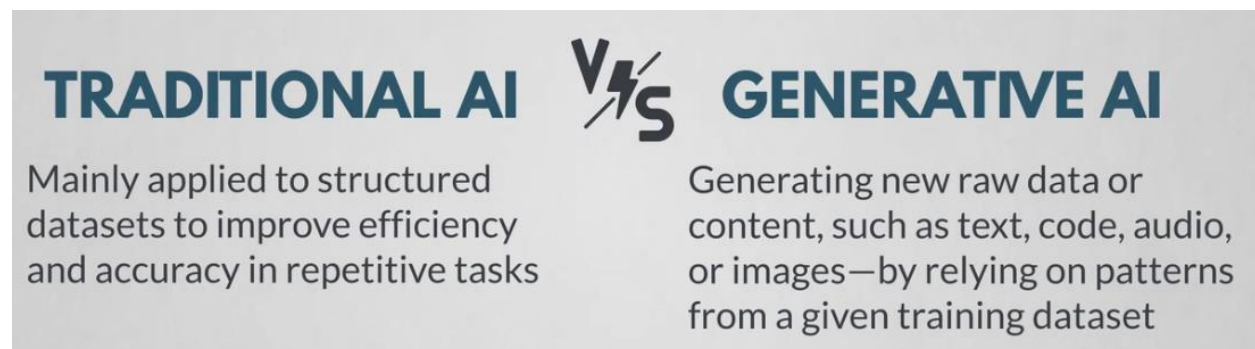
It branches off into two areas:

- 1- Qualitative (intuition + analysis)
- 2- Quantitative (formulas + algorithms)

** In practice, analytics has become a term comprising both analysis and analytics.



Business intelligence: the process of analyzing and reporting historical business data, it is the preliminary step of predictive analytics



Automated report generation: a gen AI application which is a combination of EDA and ML.

Scenario planning: a gen AI application that focuses on long-term strategies once product and feature implementation has been established.

Product Personalization: Collecting user and product data and developing quantitative predictive models.

Artificial general intelligence (AGI) is a field of theoretical AI research that attempts to create software with human-like intelligence and the ability to self-teach. The aim is for the software to be able to perform tasks that it is not necessarily trained or developed for and to work on multiple tasks rather than only a single task.

If it becomes a reality, a machine can create and use new knowledge autonomously.

**MEASURE - related to a simple
descriptive statistics of past performance**

METRIC = MEASURE + BUSINESS MEANING

**KPIs = metrics + business objectives
= Key Performance Indicators**

Examples:

metric

- the traffic of a page
from your website that was
visited by any type of user

KPI

- the traffic generated
only from users who have
clicked on a link provided
in your ad campaign

BI: Tools and techniques Measures Observations Metrics Dashboards	Traditional methods: Tools and techniques Factor analysis Logistic regression Linear regression Non-linear regression
BI: Real-life examples Price optimisation Inventory management	Traditional methods: Real-life examples Customer lifetime value prediction User experience (UX) Forecasting sales volume

Using the illustrated example, order the components of a supervised machine learning process.

1

Data

2

Model

3

Objective function

4

Optimization algorithm

Using the illustrated example, order the components of an unsupervised machine learning process.

1 Data

2 Objective function

3 Model

4 Optimization algorithm

Using the illustrated example, order the components of a reinforcement machine learning process.

1 Model

2 Objective function

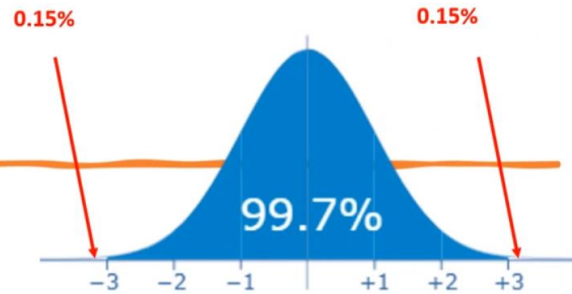
3 Data

4 Optimization algorithm

Two-Tailed Test

- $\bar{x} = 1.05$ and $s = 0.5 s$.
- $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}} = \frac{0.5}{\sqrt{100}} = 0.05$
- $z \text{ or } t = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{1.05 - 1.2}{0.05} = -3$

- If we use $z = -3$ this means 1.05 is below 1.2 by 3 standard deviations.



Type1 error: reject a true null hypothesis (false positive), the probability of making this error is α

Type2 error: accept a false null hypothesis (false negative), the probability of making this error is $1 - \beta$.

β depends mainly on sample size and magnitude of the effect.

$1 - \beta$ is called the power of the test

Decision rule:

Reject if: absolute value of Z-score > positive critical value (z)

If we asked to retest a hypothesis using a higher significance level?

There is no need for the test as we have already rejected it lower significance levels.

P-value is the smallest level of significance at which we can still reject the null hypothesis, given the observed sample statistic

You should reject the null hypothesis if $p\text{-value} < \alpha$

The probability of correctly rejecting a null hypothesis when it is false is called **power of a test**.

p-value indicates the probability of making a **type 2 error**.

- Between two categorical variables (**Chi-square test**). (*Difference between more than one proportions – nonparametric test- extension to proportion-test*).
- Between one categorical and one continuous (**ANOVA**). (*Difference between more than two means. i.e., extension to t-test*).
- Between two continuous variables (**correlation & regression analysis**).
- Obtain p-value and check for the significance of the relationship.

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Comparing Proportions (the following example will be of Bernoulli distribution):

- The sampling distribution of the sample proportion for each group is normally distributed with $\mu_{\bar{p}} = p$ and $\sigma^2_{\bar{p}} = \frac{p(1-p)}{n}$.
- The sampling distribution of the proportion difference is normally distributed with $\mu_{\bar{p}_1 - \bar{p}_2} = p_1 - p_2$ and $\sigma^2_{\bar{p}_1 - \bar{p}_2} = \sigma^2_{\bar{p}_1} + \sigma^2_{\bar{p}_2}$
- Very few real-world observations follow a chi-square distribution. The main purpose of chi-square distributions is hypothesis testing, not describing real-world distributions.
- Imagine taking a random sample of a standard normal distribution (Z). If you squared all the values in the sample, you would have the chi-square distribution with $k = 1$. $\chi^2_1 = Z^2$
- Now imagine taking samples from two standard normal distributions (Z_1 and Z_2). If each time you sampled a pair of values, you squared them and added them together, you would have the chi-square distribution with $k = 2$. $\chi^2_2 = Z_1^2 + Z_2^2$

- There are two types of Pearson's chi-square tests:

- 1) The **chi-square goodness of fit test** is used to test whether the **frequency distribution** of a categorical variable is different from your expectations. **E.g., Is a six-sided die fair?**

Side	1	2	3	4	5	6
rolls	22	23	24	25	24	32

- 2) The **chi-square test of independence** is used to test whether two categorical variables are related to each other. **E.g., Is participating in a study group related to passing an exam?**

	Failed Exam	Passed exam
Not in a study group	12	42
In a study group	6	38

For both types:

$$\chi^2 = \sum_{k=1}^n \frac{(O_k - E_k)^2}{E_k}$$

Observed frequency Expected frequency
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- **The following conditions are necessary if you want to perform a chi-square goodness of fit test:**

- You want to test a hypothesis about the distribution of one categorical variable. If your variable is continuous, you can convert it to a categorical variable by separating the observations into intervals. This process is known as **data binning**.
- The sample was randomly selected from the population.
- There are a minimum of **five observations expected** in each group.

F-Distribution

- The **F-Distribution** with d_1 and d_2 degrees of freedom is the distribution of $\frac{S_1/d_1}{S_2/d_2}$ where S_1 and S_2 are independent random variables with chi-square distributions with respective degrees of freedom d_1 and d_2 .
 - i.e., The F-distribution results from the division of two chi-square distributions which are divided by the respective degrees of freedom.
 - It is mainly used in **ANOVA** test.
 - **ANOVA has two main types of tests:**
 - One-way (Factor) ANOVA.
(mean of a **quantitative variable** changes according to the **levels** of **one** categorical variable (**factor**).)
 - Two-way (Factor) ANOVA.
(mean of a **quantitative variable** changes according to the **levels** of **two** categorical variables (**factors**).)
 - Both of them can be used for dependent (repeated measures) and independent (without repeated measures) data.
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- **So, we can say we have 4 types of ANOVA tests.**
 - **One-way (Factor) ANOVA without repeated measures.**
 - One-way (Factor) ANOVA with repeated measures.
 - Two-way (Factor) ANOVA without repeated measures.
 - Two-way (Factor) ANOVA with repeated measures.